When History Can Replace Action Chunking

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Abstract. We connect observability, incremental chunk stability, and recovery experiments on the linear challenging pair. Sufficient history permits exact noiseless one-step imitation, but restricted demonstrations need not identify that policy. We give a conditional certificate for general learners and an explicit small-gain certificate for the ridge learner. In the current noiseless sweep, history eight requires execution length eight under the chosen criterion, whereas history sixteen permits execution length one. These results separate state information from feedback selection; they do not establish universal monotonicity of tracking loss.

1 History supplies information; learning selects feedback

Consider $x_{t+1} = Ax_t + Bu_t$, $y_t = Cx_t$, and expert $u_t^* = Kx_t$, with $F = A + BK$. Restricted demonstrations can conceal recovery behavior even for stable plants and experts (Simchowitz et al., 2025); commitment can suppress destabilizing learned feedback (Zhang et al., 2025). We ask whether history reduces the need for that commitment.

The learner observes $z_t^{(r)} = \mathrm{col}(y_{t-r+1:t}, u_{t-r+1:t-1})$. Context r counts observations, prediction horizon H counts predicted actions, and execution $\ell \le H$ counts actions committed before replanning. Let $h = \min\{r : \operatorname{rank} \mathcal{O}_r = n\}$, where $\mathcal{O}_r = \mathrm{col}(C, CA, \dots, CA^{r-1})$. Reconstruction assumes noiseless measurements, known applied inputs, known dynamics, and a valid initial history.

Proposition 1 (action recovery and execution equivalence). An exact current-action policy exists on all noiseless histories iff

$$\ker \mathcal{O}_r \subseteq \ker(KA^{r-1}).$$

For $r \ge h$, state reconstruction is $x_t = R_r z_t$, with

$$R_r = [A^{r-1}\mathcal{O}_r^\dagger \quad \mathcal{R}_r - A^{r-1}\mathcal{O}_r^\dagger \mathcal{T}_r],$$

where $Y = \mathcal{O}_r x_{t-r+1} + \mathcal{T}_r U$ and $x_t = A^{r-1}x_{t-r+1} + \mathcal{R}_r U$ define the usual input maps. The chunk policy $\pi_k^*(z) = KF^k R_r z$ then reproduces the expert for every execution length $1 \le \ell \le H$.

Proof. Identical histories with identical inputs differ initially by $d \in \ker \mathcal{O}_r$; their current expert actions differ by $KA^{r-1}d$. Kernel inclusion is necessary and sufficient for action recovery. When $r \ge h$, left inversion and propagation give R_r. Applying $KF^k x_b$ produces $x_{b+k} = F^k x_b$ by induction. □

Thus observability removes a representational need for chunking, not a statistical learning difficulty. Noiseless expert histories have rank at most n; narrow histories can have rank one. More context can enlarge the set of unidentified coefficients. Under shared process disturbances, one-step tracking obeys $d_{t+1} = Fd_t + Be_t$, where $e_t = \hat{\pi}(z_t) - Kx_t$. If $\|F^k\| \le M\beta^k$ and deployment error is at most ε, then $\limsup_t \|d_t\| \le M\|B\|\varepsilon/(1-\beta)$. Training MSE alone does not supply this deployment bound.

2 Chunk stability and a conditional history tradeoff

Proposition 2 (Horton's history-based chunk guarantee). Let $\|z\|_{\mathcal{Z}}$ sum the Euclidean norms of its observation/action blocks. Suppose all admissible trajectory pairs satisfy

$$\|\delta y_{t+k}\| \le C_0 \rho^k \|\delta z_t\|_{\mathcal{Z}} + C_0 \sum_{i<k} \rho^{k-1-i} \|\delta u_{t+i}\|,$$

with $C_0 \ge 1$ and $0 < \rho < 1$. Use the same policy and schedule on both trajectories, $u_{j\ell+k} = \pi_k(z_{j\ell}) + v_{j\ell+k}$, with $\mathrm{Lip}_{\mathcal{Z}}(\pi_k) \le D\rho^k$. For $0 < \alpha < 1$, set

$$\bar{\rho} = \rho^{1-\alpha}, \quad C_r = 2rC_0\rho^{-(r-1)}, \quad \bar{C} = C_0\left(1 + \frac{D}{e\alpha\rho\log(1/\rho)}\right).$$

If $\ell \ge \max(1, r-1)$ and $C_r(1 + D\ell/\rho)\rho^{\alpha\ell} \le 1$, then

$$\|\delta y_t\| \le \bar{C}\bar{\rho}^t \|\delta z_0\|_{\mathcal{Z}} + \bar{C}C_r \sum_{i<t} \bar{\rho}^{t-1-i} \|\delta v_i\|.$$

Proof sketch. With $Z_j = \|\delta z_{j\ell}\|_{\mathcal{Z}}$ and $V_i = \|\delta v_i\|$, substitute $\|\delta u_{j\ell+k}\| \le D\rho^k Z_j + V_{j\ell+k}$ into the assumed bound. Summing the blocks of the next history gives

$$Z_{j+1} \le C_r(1 + D\ell/\rho)\rho^\ell Z_j + C_r \sum_{i<\ell} \rho^{\ell-1-i} V_{j\ell+i}.$$

The first coefficient is at most $\bar{\rho}^\ell$. Iterate and use $k\rho^{\alpha k} \le [e\alpha\log(1/\rho)]^{-1}$ within chunks. □

For observable Schur-stable linear plants with $r \ge h$, reconstruction yields the assumed output bound, with suitable constants. This completes the supplied draft's argument, but *does not prove that context shortens execution*: C_r grows with r, and $\ell \ge r-1$ cannot certify $r = 16, \ell = 1$.

Proposition 3 (conditional certificate for general learners). Choose a norm with $\|F\| \le a < 1$. Suppose an arbitrary deterministic learner satisfies, on every admissible deployment history,

$$\left\|\hat{\pi}_{r,j}(z_t) - KF^j x_t\right\| \le \gamma_r q^j \|x_t\|, \qquad 0 < q < 1.$$

A sufficient condition for exponential stability is

$$G(r,\ell) := a^\ell + \gamma_r \sum_{j<\ell} \left\|A^{\ell-1-j}B\right\| q^j < 1.$$

Proof. Writing the prediction residual as e_j, unrolling gives $x_{t+\ell} = F^\ell x_t + \sum_{j<\ell} A^{\ell-1-j}Be_j$; hence $\|x_{t+\ell}\| \le G(r,\ell)\|x_t\|$. Finite within-chunk gains extend decay to all phases. □

If γ_r decreases, certified feasible execution sets expand; their shortest tested length cannot increase. In particular $\gamma_r \|B\| < 1 - a$ certifies $\ell = 1$. If A is also stable and the envelope extends to arbitrary H, $G(r,\ell) \to 0$ as $\ell \to \infty$. The relative-error assumption is strong and is *not established* for our raw-history ridge learner. Neither observability nor additional features guarantees decreasing deployment error. The certificate concerns stability, not monotone tracking loss.

3 A specific mechanism for the narrow-data ridge learner

For the challenging pair at $\mu = 1/8$, our experiment fixes

$$A_1 = \begin{bmatrix} 1+\mu & c \\ -c & 1-2\mu \end{bmatrix}, \quad K_1 = \begin{bmatrix} -(1+\mu) & -c \\ c & 0 \end{bmatrix}, \quad c = 3\mu/2.$$

Both A_1 and A_1+K_1 are stable. On the demonstration support $x_1 = 0$, $x_{2,t+1} = qx_{2,t}$ and $u_{1,t} = -cx_{2,t}$, with $q = 3/4$ and $c = 3/16$. Every feature is $x_{2,t}v_r$: its observation coefficients are q^{-k} in the x_2 slots, $0 \le k < r$, and past-action coefficients are $-cq^{-k}$ in the u_1 slots, $1 \le k < r$; all other slots are zero.

Let $S = \sum_i x_{2,t_i}^2 > 0$, $\delta = \lambda/S$, and

$$D_r = \|v_r\|_2^2 = 1 + (1+c^2)\sum_{k=1}^{r-1} q^{-2k}, \qquad b_H = \mathrm{col}((-cq^j, 0))_{j=0}^{H-1}.$$

Ridge regression minimizes $\|XW^\top - Y\|_F^2 + \lambda\|W\|_F^2$. Since $X = sv_r^\top$ and $Y = sb_H^\top$, its normal equations give exactly

$$\widehat{W} = \frac{b_H v_r^\top}{D_r + \delta}.$$

Thus one-step execution uses

$$u_{1,t} = -\frac{c}{D_r+\delta}\sum_{k=0}^{r-1} q^{-k}x_{2,t-k} + \frac{c^2}{D_r+\delta}\sum_{k=1}^{r-1} q^{-k}u_{1,t-k}, \qquad u_{2,t} = 0.$$

History changes the ridge-selected feedback outside the demonstrations; it does not increase their rank or identify the missing expert correction to x_1.

Proposition 4 (history can stabilize one-step ridge control). Define

$$\alpha_r = \frac{c\sum_{k=0}^{r-1} q^{-k}}{D_r+\delta}, \qquad \beta_r = \frac{c^2\sum_{k=1}^{r-1} q^{-k}}{D_r+\delta}.$$

The one-step controller is asymptotically stable on A_1 if

$$48\alpha_r + \beta_r < 1.$$

Proof. Write $A_1 = aI + N$, $a = 15/16$, where $N^2 = 0$ and $N_{21} = -c$. The absolute impulse-response gain from u_1 to x_2 is

$$\sum_{k\ge 0} |\mathbf{e}_2^\top A_1^k \mathbf{e}_1| = c\sum_{k\ge 1} ka^{k-1} = \frac{c}{(1-a)^2} = 48.$$

The observation and autoregressive filter coefficient sums are α_r and β_r. Their combined strictly causal feedback gain is at most $48\alpha_r + \beta_r$; a gain below one gives a stable inverse by the Neumann series. Remaining plant and buffer modes are stable. □

Because $D_r = \Theta(q^{-2r})$, both gains are $O(q^r)$: sufficiently long history stabilizes one-step execution. At $\delta = 0$, the certificate is 4.9066, 1.8536, and 0.2021 for $r = 4, 8, 16$. Hence $r = 16$ has an analytic certificate; failing the sufficient test at $r = 8$ proves nothing by itself. At fixed r, predicted chunks satisfy $\hat{u}_{t+j|t} = q^j\hat{u}_{t|t}$; if extended indefinitely, stable plant dynamics and decaying committed actions also yield eventual chunk-boundary contraction. Neither argument orders every finite-horizon loss. This mechanism depends on the stable plant, feature scaling, and ridge selection, not observability alone.

4 Evidence, interpretation, and limits

The retained midterm experiment varies partial observation on stable cyclic plants. For $n = 10, h = 4$, context three with one-step execution fails despite near-zero validation error; two-step execution stabilizes it. Context four permits one-step execution with mean tail MSE 7.64×10^{-6}, versus 2.80×10^{-4} for context three and execution two. These earlier noisy results illustrate action-relevant history; they were not rerun here.

The current full-observation sweep ($h = 1$) trains 375 configurations using 32 narrow noiseless demonstrations per seed, five seeds, $\lambda = 10^{-6}$, and matched target times across r, H. All controllers share 20 held-out initial-state pairs: x_2 uniform on $[-1, 1]$, with $x_1 = 0$ or a signed departure of magnitude 0.0005–0.0015. After a common 15-step expert history, the departure is injected; there is no continuing noise. Test MSE averages full-state tracking error over 200 post-action states and the 20 perturbed rollouts. Separate 1000-step sample losses diagnose long-term growth.

History r	Execution ℓ	Median test MSE	Worst per-step rate
4	16	1.379×10^{-5}	1.00548
8	1	7.396×10^{-4}	1.01892
8	8	2.260×10^{-6}	0.99544
16	1	6.487×10^{-7}	0.97110

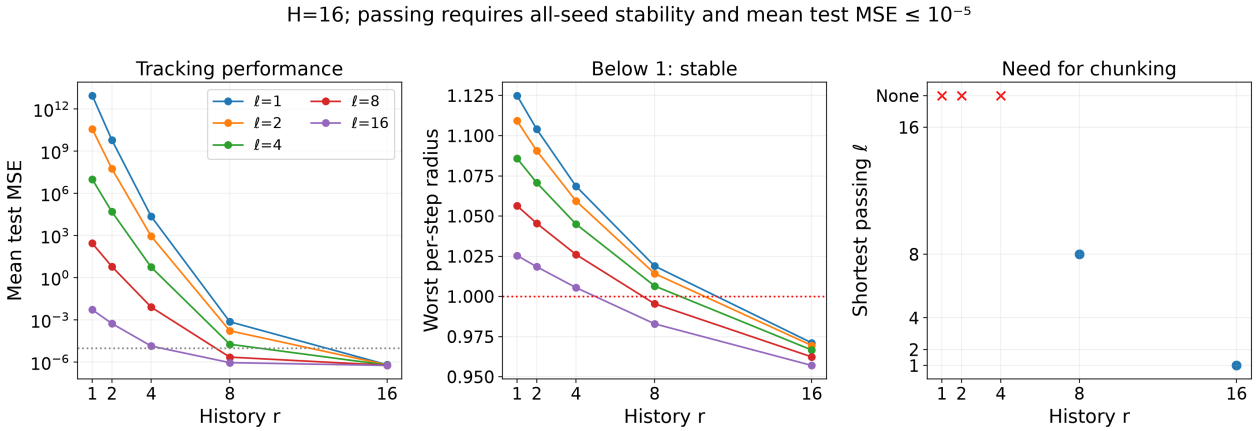


Figure 1: At $H = 16$, history improves test loss and spectral stability. Shortest passing execution is 8 at $r = 8$ and 1 at $r = 16$; None means no tested setting passes. Passing requires stability and mean test MSE $\leq 10^{-5}$ for every seed.

Conclusion. The current $r = 16, \ell = 1$ loss is below zero control's 0.08217, but above the model-informed memory witness's 1.736×10^{-8}. Since $h = 1$, this is feedback selection rather than hidden-state recovery. The earlier negative narrow-data result used continuing noise and $r \leq 8$; it is a different protocol. The threshold is illustrative and five seeds give descriptive evidence. Generalizing the tradeoff requires proving that deployment error decreases with context; observability and training fit do not suffice. No result here establishes universal monotonicity of loss.

References

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